

Evolving Offensive Football Strategies Using Genetic Algorithms

CS 523 Final Report — Spring 2026

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Abstract

This project investigates whether emergent offensive play calling strategies arise when a genetic algorithm evolves strategy parameters in a stochastic football environment, and what those emergent strategies look like. Each strategy is encoded as a real valued genome with four parameters: run probability, short pass probability, deep pass probability, and fourth down aggression. Strategies were evaluated using a custom drive simulator calibrated to over 106,000 real NFL plays from the 2022-2024 seasons via the nflfastR play-by-play dataset. A genetic algorithm with tournament selection, uniform crossover, Gaussian mutation, and elitism evolved a population of 100 strategies over 100 generations across six experimental conditions, with the base fitness function defined as $mean(points/drive)$. Subsequent runs progressively refined either the fitness function itself, adding a variance penalty, or the simulator’s drive scoring rules, adding turnover penalties, situational constraints, or field-position-adjusted EPA values. All experiments were repeated across 10 fixed random seeds for statistical stability across runs. Results show that emergent strategy is determined entirely by the fitness landscape: with no turnover penalty (Runs 1, 2, 3), the GA consistently converged to deep pass dominant strategies, approximately 94% deep, regardless of variance penalties or situational constraints, suggesting the NFL data itself contains a single steep fitness peak. Adding a -3.5 point Expected Points Added (EPA) turnover penalty completely inverted the emergent behavior (Runs 4a and 4b), with short pass dominant strategies emerging at 82–93%. Run 5, the most refined condition combining the turnover penalty with field-position-adjusted EPA, revealed a bimodal landscape where deep and short pass strategies coexisted as competing peaks. Across all 60 GA runs, fourth down aggression universally converged to approximately 90% or greater, an emergent finding consistent with real world NFL analytical trends toward more aggressive fourth down decision making. These results show that it’s the structure of the fitness function that shapes emergent behavior in a complex adaptive system, and that genetic algorithms can independently discover strategic patterns found within raw sports data.

1 Introduction

Football play calling is a complex adaptive decision making problem in which coaches must repeatedly select among run, short pass, and deep pass options under uncertainty, balancing risk against reward across constantly changing game states. Mitchell (2009) frames **complex adaptive systems (CAS)** as agents whose local interactions produce **emergent system level behaviors** not

predictable from individual components, a definition that fits offensive play calling, where individual play decisions interact stochastically to produce drive level outcomes. Prior work has looked at play calling strategies through several methodologies: Romer (2006) used dynamic programming to show NFL coaches are too conservative on fourth down, Biro and Walker (2021) modeled play calling as a Markov decision process solved via reinforcement learning, and Fernández Leiva, Cotta and Ceballos (2008) evolved emergent team strategies using genetic algorithms in simulated football environments. This project asks: *Does a genetic algorithm produce emergent offensive play calling strategies when evolving against real NFL data, and if so, what strategies emerge under different fitness conditions?* A **genetic algorithm (GA)** is an evolutionary search procedure that uses selection, crossover, and mutation to optimize a population of candidate solutions over successive generations. Following the GA framework provided in the course README_GA.md (Moses, 2026), each strategy is encoded as a real valued genome of four parameters (run probability, short pass probability, deep pass probability, and fourth down aggression), with fitness defined as average points per drive across a drive simulator. Six experimental conditions were tested across 10 fixed random seeds each, progressively refining either the fitness function or the simulator’s drive scoring rules to investigate how environmental pressure shapes emergent behavior, the central concern of complex adaptive systems analysis.

2 Methods

2.1 Code Generation

The simulation framework was implemented in Python and consists of four components: a data acquisition module that pulls NFL play-by-play data and computes empirical probability distributions, a drive simulator that models a single offensive possession, a genetic algorithm that evolves a population of strategies, and an experiment runner that orchestrates multi-seed evaluations. The structural framework for the genetic algorithm follows the README_GA reference provided for the course (Moses, 2026), and code was generated using Claude Code (see Section 2.3) based on detailed specifications, design decisions, and iterative refinements provided throughout development.

2.1.1 Data Acquisition and Probability Distributions

NFL play-by-play data for the 2022–2024 seasons was acquired using the `nfl_data_py` Python library, which provides a Python interface to the `nflfastR` dataset. After filtering to scrimmage plays only (non-kickoff, non-extra-point plays where the offense ran an actual play from scrimmage), the dataset contained over 106,000 plays. Empirical probability distributions were computed separately for three play types: run plays, short passes (passes traveling less than 20 yards through the air), and deep passes (passes traveling 20 or more yards through the air). For each play type, the following statistics were extracted directly from the data: the full empirical distribution of yards gained, completion rates (for pass plays), interception rates, fumble lost rates, and touchdown rates. These distributions form the outcome engine for the simulator: when a play is called during simulation, the result is sampled directly from the corresponding real NFL distribution rather than from assumed or hand tuned probabilities. This puts every simulated outcome in

observed NFL behavior, with defensive resistance encoded in the empirical distributions, since the data was generated from real games where defenses were actively trying to stop the offense.

2.1.2 Drive Simulator

The drive simulator models a single offensive possession by tracking four state variables on every play: down (1–4), yards to go for a first down, field position (yardline 0–100, where 0 is the offense’s own end zone and 100 is the opponent’s end zone), and points scored. On each play, the strategy’s genome parameters are used to select a play type by sampling from a weighted probability distribution over run, short pass, and deep pass. After a play type is selected, the outcome is simulated by sampling from the corresponding NFL distribution that was derived. Turnovers (interceptions and fumbles) and completions are checked stochastically against the real data probabilities before yardage is applied. After each play, the simulator updates field position, checks for touchdowns ($yardline \geq 100$, scoring 7 points), a turnover, first downs (resetting to 1st and 10), and 4th down decisions. On 4th down, the simulator either attempts a field goal ($yardline \geq 67$, scoring 3 points), goes for it (governed by the strategy’s *fourth_down_aggression* parameter), or punts. The drive terminates with a point value of 7 for a touchdown, 3 for a successful field goal, or 0 for a punt, turnover, or failed 4th down conversion. A safety cap of 50 plays was implemented to prevent runaway drives.

2.1.3 Genetic Algorithm

A genetic algorithm evolves a population of 100 offensive strategies over 100 generations. Each strategy is represented as a real valued genome with four parameters: *run_prob*, *short_pass_prob*, *deep_pass_prob*, and *fourth_down_aggression*. The three play probabilities are normalized to sum to 1.0 after every genetic operation. Fitness is evaluated by running each strategy through 100 simulated drives and computing the resulting score according to the experimental condition (see Section 2.1.4). Selection uses tournament selection with $k = 5$: five strategies are randomly sampled from the population, and the fittest is selected as a parent. Crossover is uniform across the four genome parameters: for each parameter, the child independently inherits from one of the two parents. After crossover, the three play probabilities are renormalized to sum to 1.0. Mutation applies Gaussian noise with mean 0 and standard deviation 0.05 to each parameter, with values clipped to the range $[0, 1]$ before renormalizing. Elitism preserves the top 5 strategies in each generation, helping prevent fitness regression. The framework for the GA implementation follows the course README_GA (Moses, 2026), which highlights tournament selection, uniform crossover, Gaussian mutation, and elitism as the core concepts used here.

2.1.4 Experimental Design

Six experimental conditions were tested. Run 1 used the baseline fitness function $f = \text{mean}(\text{points}/\text{drive})$ with no penalties or constraints. Run 2 added a variance penalty: $f = \text{mean}(\text{points}/\text{drive}) - 0.5 \times \text{std}(\text{points}/\text{drive})$, in an attempt to penalize strategies whose drive scores were highly inconsistent. Run 3 retained the variance penalty and added situational play calling constraints inside the simulator: deep pass probability was capped at 20% on 3rd or 4th down with three or fewer yards to go, deep pass probability was capped at 30% when the offense was inside their own 20 yard line,

and short pass probability received a $+0.20$ boost when yards to go exceeded 15. Run 4a applied a -3.5 point Expected Points Added (EPA) penalty whenever a turnover (interception or fumble) occurred, while keeping the fitness function as $f = \text{mean}(\text{points}/\text{drive})$. Run 4b combined the variance penalty with the turnover penalty. Run 5 added field-position-adjusted EPA values to punts and failed 4th down conversions: punts in own territory received small negative penalties (-0.2 to -1.0), failed 4th downs received larger negative penalties (-0.3 to -2.0), and punts in opponent territory received zero. The fitness function itself remained $f = \text{mean}(\text{points}/\text{drive})$ across Runs 1, 3 (with constraints applied to drive scoring), 4a, and 5; only Runs 2 and 4b modified the fitness function formula by introducing the variance penalty. All other experimental refinements (turnover penalties, situational constraints, field-position EPA) modified how individual drive scores were computed within the simulator rather than the fitness aggregation itself. Each of the six experimental conditions was run across the same set of 10 fixed random seeds to ensure that observed differences between runs were attributable to the experimental change and not to seed variance. For each run, fitness over generations was tracked by recording the best, mean, and worst fitness values in the population at every generation across all 10 seeds.

2.2 Code Testing and Verification

Each component of the simulation was tested independently before integration, and outputs were validated against real NFL behavior throughout development. The data acquisition module was verified by checking the computed probability distributions against published NFL averages: run plays averaged 4.5 yards with a 0.67% fumble rate, short passes completed at 70% with a 1.5% interception rate, and deep passes completed at 41.8% with a 5.8% interception rate, all consistent with the 2022–2024 seasons (Pro-Football-Reference, 2024). The drive simulator was tested by running drives in verbose mode with a fixed strategy and manually verifying down and distance tracking, first down resets, touchdown triggers, 4th down decisions, and edge cases such as drives ending in own end zone or consecutive turnovers. The genetic algorithm was verified to preserve the standard GA invariants after every crossover and mutation operation (probabilities summing to 1.0, parameters bounded in $[0, 1]$, top 5 strategies preserved via elitism).

2.3 AI Usage

Claude Code (Anthropic, used March–May 2026) was used to generate portions of the code in this project, working from specifications and design decisions that I provided throughout development. Significant portions of the drive simulator were done directly by me, with Claude Code primarily handling the data acquisition module, the genetic algorithm implementation, figure creation and the experiment runner. I made a project brief document (`CLAUDE.md`) to define the genome, fitness function variants, simulator state tracking, situational constraints, GA parameters, and required output figures, which Claude Code read at the start of each session to maintain the context of the project in development. Claude Code was also used for insight and debugging help during hypothesis refinement and result interpretation. I used it to compare possible explanations for why deep pass dominance emerged in early runs. All code outputs from Claude Code were validated by myself through the tests described in Section 2.2, and all results were verified and I chose the final interpretations and refinements to proceed with.

3 Results

All six experimental conditions were evaluated across the same set of 10 fixed random seeds, producing 60 total genetic algorithm runs. The results reveal three distinct fitness landscape regimes corresponding to three different sets of experimental conditions. Runs 1, 2, and 3, which omit any turnover penalty, converged to deep pass dominant strategies regardless of variance penalties or situational constraints. Runs 4a and 4b, which introduced a -3.5 EPA turnover penalty, completely inverted the emergent strategy toward short pass dominance. Run 5, which combined the turnover penalty with field-position-adjusted EPA values for punts and failed 4th down conversions, revealed a bimodal fitness landscape where deep pass and short pass strategies coexisted as competing peaks. Across all 60 runs, fourth down aggression universally converged to 0.91 or higher, an emergent finding. The following three subsections present each regime in detail.

3.1 Regime 1: Deep Pass Dominance Without Turnover Penalty (Runs 1, 2, 3)

Runs 1, 2, and 3 progressively refined the fitness function and simulator constraints without introducing any penalty for turnovers. Run 1 used raw mean fitness, Run 2 added a variance penalty ($\text{mean} - 0.5 \times \text{std}$), and Run 3 retained the variance penalty while adding situational play calling constraints. Despite these refinements, all three runs converged to the same emergent strategy: deep pass dominance at 90.4–90.9% across the population.



Figure 1: Fitness over generations for Run 1 (raw mean fitness, no penalties or constraints). Mean best fitness across 10 fixed seeds with ± 1 standard deviation shading.

A notable feature of the fitness curves across all three runs is the rapid plateau after approximately generation 10-15, with little improvement in subsequent generations. Fitness rises rapidly during the first 10 generations and plateaus near 4.3 points per drive thereafter, with Runs 2 and 3 producing similar plateau patterns but at lower absolute fitness values due to the variance penalty. This reflects premature convergence, an occurrence in genetic algorithms where the population loses diversity once it locates a strong fitness peak (Sudholt, 2018). With elitism preserving the top 5 strategies and tournament selection rewarding fit individuals, the population collapses around the dominant strategy, after which only small Gaussian mutations drive further changes. Because the fitness landscape contains a single dominant peak in these runs, the GA reaches it quickly and climbs around it without finding any meaningfully better solutions. This is the expected behavior in a single peak fitness landscape.

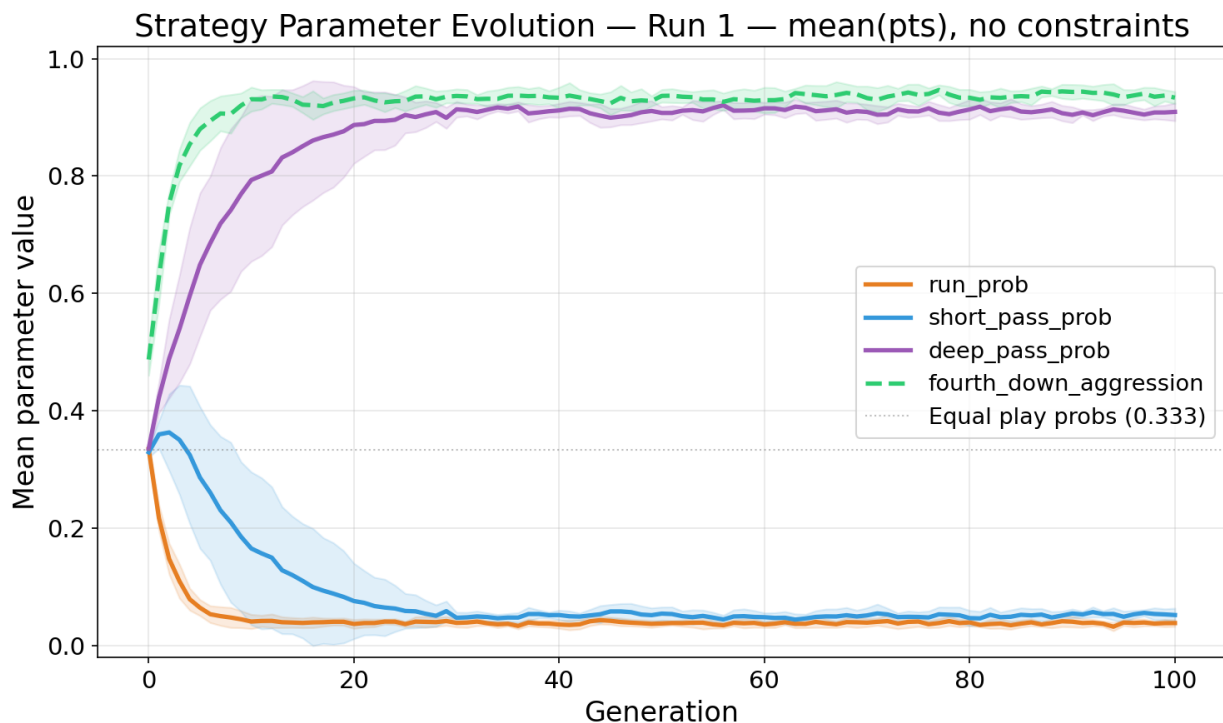


Figure 2: Strategy parameter evolution for Run 1. Population means across 10 fixed seeds with ± 1 standard deviation shading.

The persistence of deep pass dominance across all three runs reveals that the empirical NFL data contains a single dominant fitness peak corresponding to a deep pass heavy, aggressive strategy. Deep pass probability (purple) converges to 91% by generation 20 while run (orange) and short pass (blue) collapse to below 6%, with fourth down aggression (green dashed) converging to 0.93. Runs 2 and 3 produced nearly identical parameter evolution trajectories despite the variance penalty and situational constraints, indicating these refinements affected fitness magnitude but not the emergent strategy itself. The deep pass play type produces 28.2 yard average completions with a 6.9% touchdown rate, generating scoring outcomes that outweigh its 5.84% interception risk when interceptions carry no point penalty. Neither the variance penalty in Run 2 nor the situational play calling constraints in Run 3 were sufficient to change this peak in a notable way, suggesting

the fundamental issue lies in how turnovers are scored rather than in how fitness is changed across drives.

3.2 Regime 2: Strategy Inversion Under Turnover Penalty (Runs 4a, 4b)

Runs 4a and 4b introduced a -3.5 point EPA penalty whenever a turnover (interception or fumble) occurred during a drive. Run 4a kept the fitness function as raw mean points, while Run 4b combined the turnover penalty with the variance penalty from Run 2. Both runs produced a dramatic swap of the emergent strategy: short pass replaced deep pass as the dominant play type, and population means shifted from 91% deep in Regime 1 to 80% short in Run 4a and 90% short in Run 4b.

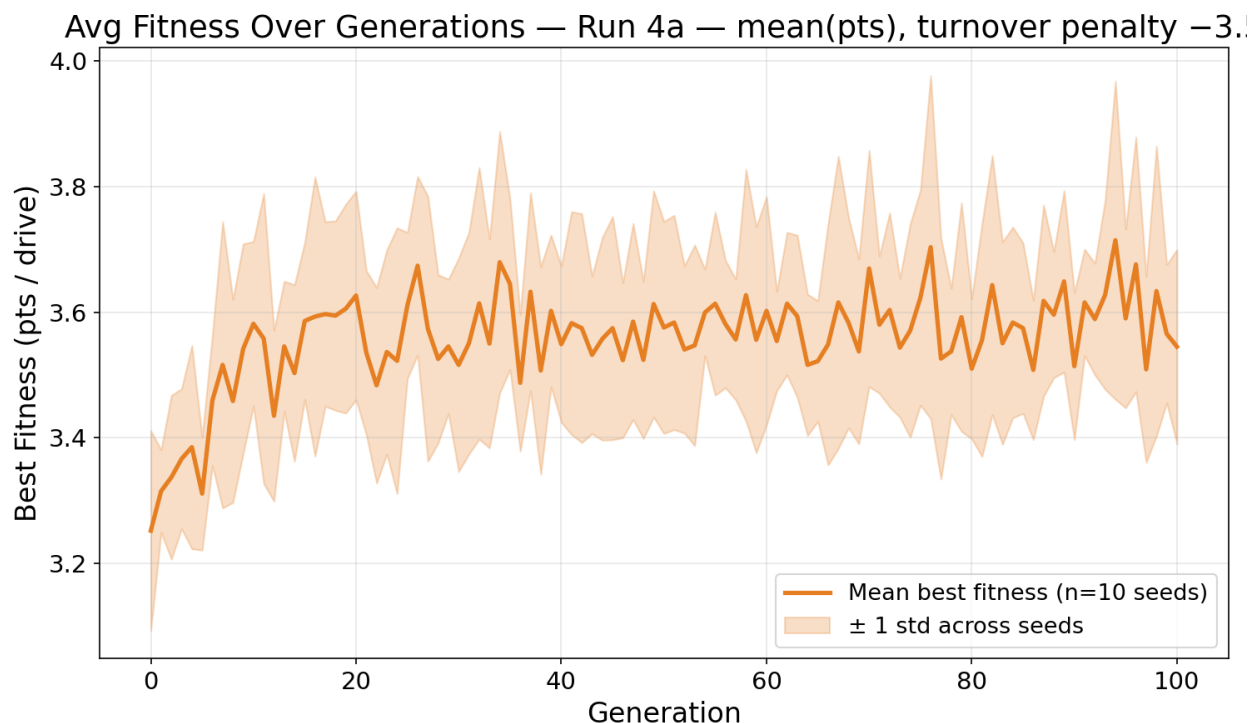


Figure 3: Fitness over generations for Run 4a (raw mean fitness with -3.5 turnover penalty). Mean best fitness across 10 fixed seeds with ± 1 standard deviation shading.

Run 4a converged to a mean best fitness of 3.545 ± 0.155 and Run 4b to 1.739 ± 0.143 , with the lower Run 4b fitness reflecting the additional variance penalty applied on top of the turnover cost. The fitness curves rise more gradually than in Regime 1, with meaningful improvement continuing through approximately generation 30 before plateauing. This slower convergence reflects a more complex fitness landscape: the GA must learn to balance the upside of completed deep passes against the high cost of interceptions, rather than simply maximizing scoring upside.

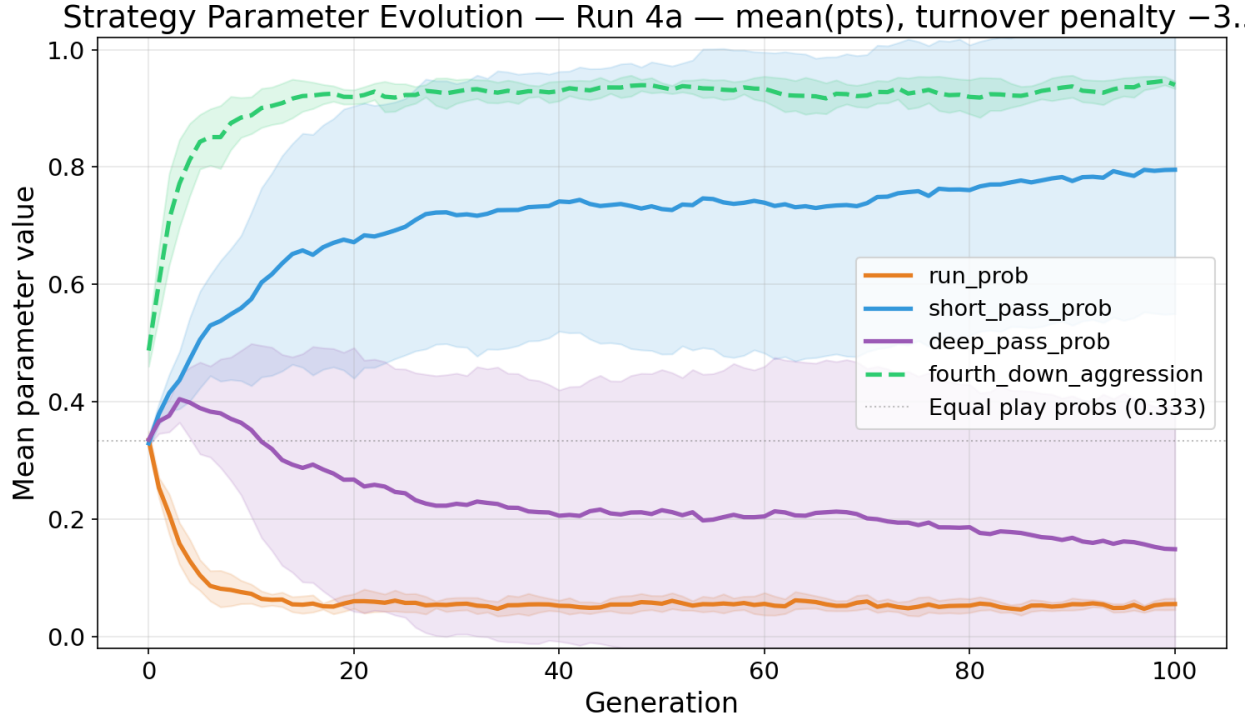


Figure 4: Strategy parameter evolution for Run 4a. Population means across 10 fixed seeds with ± 1 standard deviation shading.

A particularly notable feature of Run 4a is the wide standard deviation bands surrounding *short_pass_prob* and *deep_pass_prob*, which reflect bimodality across seeds: some seeds converged to short pass dominance (above 90%) while others converged to deep pass dominance (around 80%). This indicates the fitness landscape contains two competing peaks rather than a single dominant solution. The deep pass peak survived because completed deep passes still produce 28.2 yard average gains and frequent touchdowns, partially offsetting the turnover cost on certain seeds. Run 4b, which added the variance penalty on top of the turnover penalty, produced much narrower bands and uniform convergence to short pass dominance. This shows the variance penalty effectively eliminated the deep pass peak by additionally punishing the inconsistency that high variance deep passing creates. Across both runs, fourth down aggression converged to approximately 0.92–0.94, reflecting the universal finding from Regime 1 that going for it on fourth down is consistently better for fitness than punting in this simulation.

3.3 Regime 3: Bimodal Landscape Under Combined Penalties (Run 5)

Run 5 combined the -3.5 EPA turnover penalty from Regime 2 with field-position-adjusted EPA values applied to punts and failed 4th down conversions. This was the most realistic experimental condition, modeling not just the cost of turnovers but also the opponent’s expected scoring opportunity from field position after a punt or failed conversion. Run 5 converged to a population mean of 75.8% deep pass and 19.2% short pass, with a mean best fitness of 3.424 ± 0.214 .

Avg Fitness Over Generations — Run 5 — mean(pts), turnover + field-position

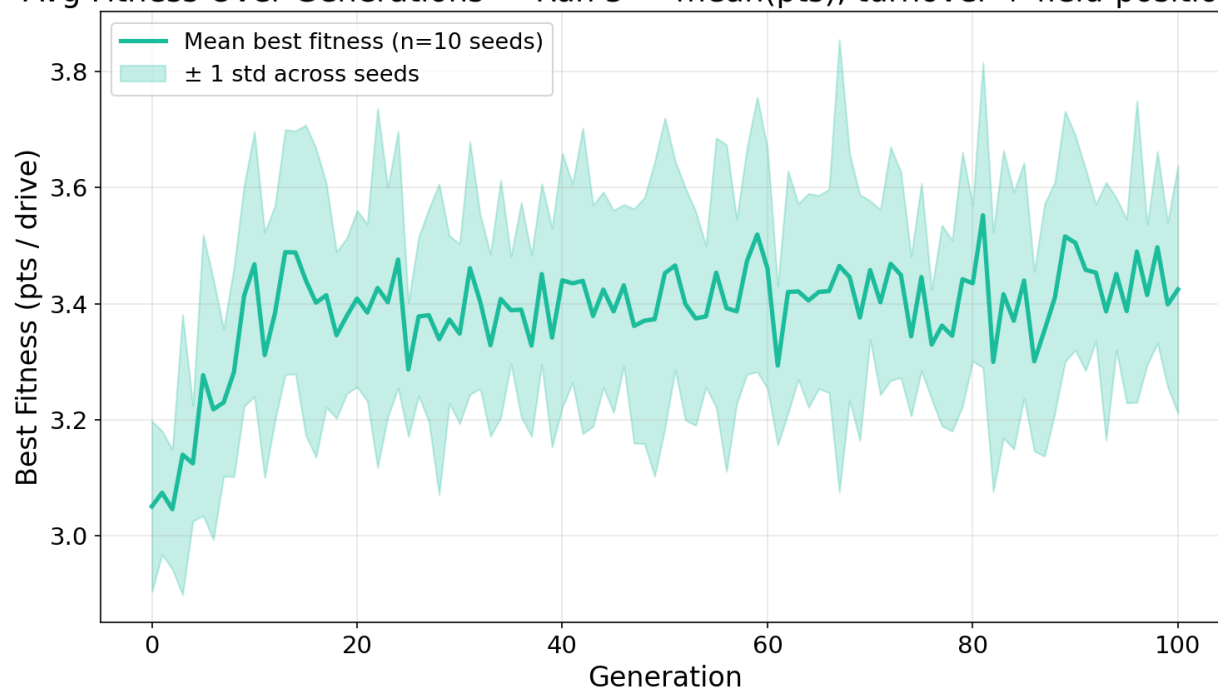


Figure 5: Fitness over generations for Run 5 (raw mean fitness with -3.5 turnover penalty and field-position-adjusted EPA for punts and failed 4th down conversions). Mean best fitness across 10 fixed seeds with ± 1 standard deviation shading.

The fitness curve plateaus at approximately 3.4 pts/drive by generation 15-20, similar to the convergence timing seen in Regimes 1 and 2. However, the curve is notably noisier than previous runs, with substantial fluctuation around the plateau value, and the standard deviation band remains consistently wide across all generations. The wide variance band reflects that different seeds converged to different local optima rather than all settling on the same fitness peak. The mean best fitness of 3.424 sits between Run 4a (3.545) and Run 4b (1.739), reflecting the additional cost imposed by negative EPA values on punts and failed 4th downs in own territory.

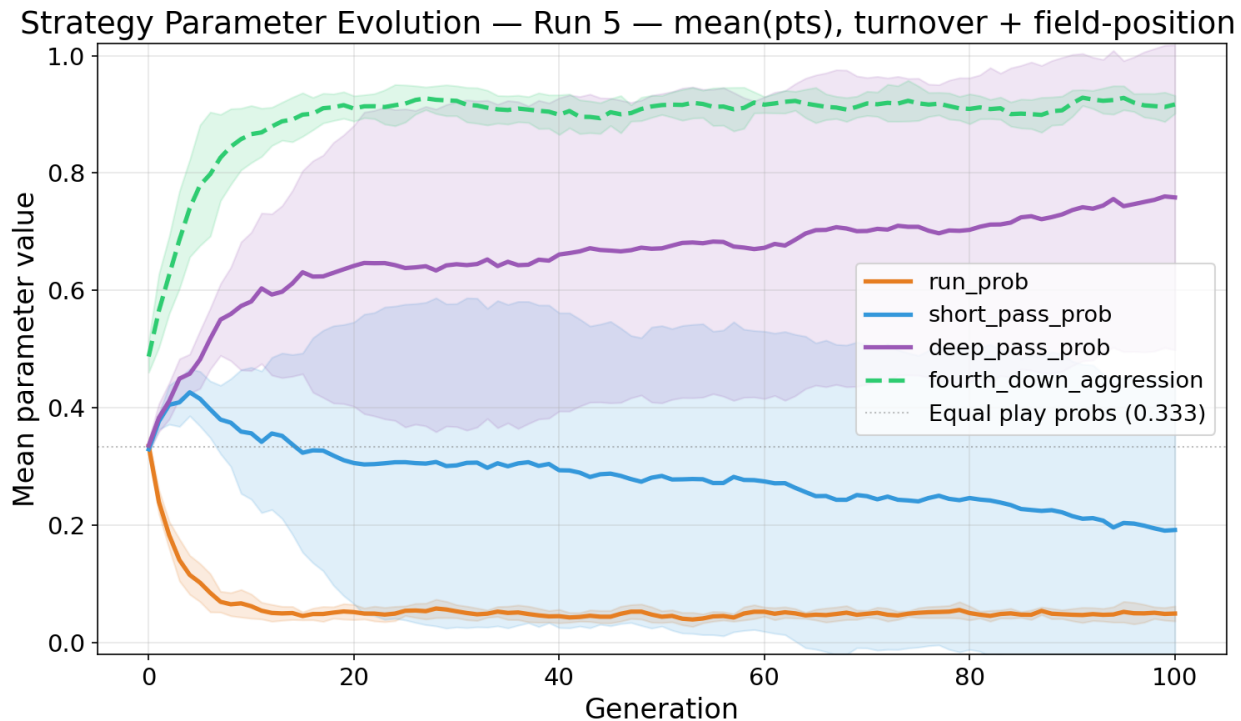


Figure 6: Strategy parameter evolution for Run 5. Population means across 10 fixed seeds with ± 1 standard deviation shading.

The wide standard deviation bands surrounding both *deep_pass_prob* and *short_pass_prob* in Figure 6 clearly visualize the bimodal fitness landscape: while the population mean settles at 75.8% deep pass, individual seeds converged across a wide range of strategies, with some finding deep pass dominant peaks and others finding short pass dominant peaks. This bimodality is a direct consequence of the field position EPA values partially offsetting the turnover penalty. When a drive ends in opponent territory through a punt (EPA = 0) or a failed 4th down (EPA = -0.1 to -0.5), the cost is much smaller than the -3.5 turnover penalty alone, making deep passes that reach opponent territory before stalling competitive again. Short passes still produce lower variance scoring drives with much lower interception risk, but they punt or fail more often in own territory where penalties are larger (-0.5 to -2.0). The result is two key strategic regions in the landscape: a deep pass strategy that exploits opponent territory cost reductions, and a short pass strategy that minimizes turnover risk while accepting smaller per play gains. Fourth down aggression again converged to approximately 0.92, consistent with every other run, confirming this finding is robust across all fitness conditions.

4 Conclusion

The key insight of this project is that emergence in this complex adaptive system is not a property of the genetic algorithm or the NFL data alone, but of their interaction through the fitness function. Across 60 total GA runs spanning six experimental conditions, the GA consistently produced emergent strategies, but which strategy emerged depended entirely on the fitness function: deep

pass dominance under raw scoring, short pass dominance under turnover penalties, and bimodal landscapes when both turnover and field position penalties were applied. Changing the rules of the environment changes what adapts, which is the defining feature of a CAS. The refinements of fitness function variants revealed three unique landscape regimes that would have been invisible from any single experimental condition, showing that experimental design itself can become a tool for refining the structure of an adaptive system.

There are two additions in the future that would advance this work. First, score differential awareness should be added to the genome, allowing fourth down aggression to adjust based on whether the team is trailing or leading. This addresses the unrealistic finding that aggression converged to the near maximum across all runs, which reflects the absence of game level context in the drive simulation. Second, a coevolutionary defensive genome should be added, replacing the implicit defensive resistance in the empirical NFL distributions with an evolved defensive strategy that adapts against the offense. This would shift the simulation from a single agent problem into a true multi agent complex adaptive system, allowing investigation of how offensive and defensive strategies coevolve under competition.

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